

HIMNISH ROUT

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Education

Bennett University

B.Tech, Computer Science and Engineering (CGPA: 7.71)

Greater Noida, India
Aug 2023 – May 2027 (Expected)

Technical Skills

- **Programming** : Python, C++, HTML, CSS, PHP, Javascript
- **Core CS** : Data Structures & Algorithms, OOP, Operating Systems
- **AI, Machine Learning & NLP**: Machine Learning, Deep Learning (CNN, RNN, Random Forest), Generative AI, LLMs, RAG Pipelines, Prompt Engineering, Vector Search, Embeddings, Fine-Tuning (LoRA/PEFT)
- **Frameworks & Libraries**: PyTorch, Scikit-learn, XGBoost, LightGBM, Hugging Face Transformers, LangChain, Pandas, NumPy, SHAP, FastAPI
- **Backend, Databases & Tools**: PostgreSQL, Git, GitHub, VS Code, Xcode

Experience

Balson Polyplast Pvt. Ltd. — IT & AI Department

AI Executive Trainee (Intern)

Ghaziabad, India (On-site)
Feb 2026 – Jun 2026

- Built a Random Forest Regressor (Scikit-learn) to predict irrigation requirements from environmental variables (moisture, temperature, humidity), cleaning and standardizing agricultural datasets with Pandas.
- Sourced and engineered training data via targeted web scraping and live sensor feeds, replacing static historical data with real-time inputs.
- Developed the machine learning prediction backend to process live soil-moisture readings, enabling real-time automation of relay-controlled irrigation valves.
- Built the web dashboard used by farmers to monitor and schedule irrigation remotely, executing full-stack development across the HTML/CSS front end and PHP/JavaScript backend.
- Collaborated cross-functionally with hardware engineers to seamlessly integrate ML models with physical hardware components in a live manufacturing environment.

Projects

LexRAG — Verified RAG Engine for Indian Legal & Financial Documents

Python, LangChain, pgvector, FastAPI, Hugging Face

- Built an end-to-end retrieval-augmented generation pipeline over RBI circulars and GST documents, combining hybrid (keyword + dense) search for higher retrieval accuracy on dense regulatory text.
- Designed a citation-verification layer so every generated answer is traceable to its source passage, effectively cutting hallucinated responses.
- Served the pipeline through a FastAPI backend with vector search for low-latency semantic retrieval over a large corpus.
- Evaluated the system against a hand-built set of legal/financial queries, measuring retrieval precision and answer faithfulness.

PayGuard — Real-Time UPI-Style Fraud Risk Scoring Engine

Python, Scikit-learn, LightGBM, XGBoost, FastAPI, Docker

- Built an end-to-end fraud detection pipeline on **6.36M mobile-money transactions** (0.08% fraud rate), covering feature engineering, class-imbalance handling, probability calibration, cost-sensitive thresholding, and a containerized **FastAPI inference service** achieving **0.71 ms p95 latency**.
- Evaluated **LightGBM, XGBoost, and scikit-learn Gradient Boosting** under identical time-based **55/15/15/15 train/validation/calibration/test splits**; on a **954K-row held-out test set**, LightGBM trained in **20.4 s (70× faster than scikit-learn)** and produced **1,264 false positives, 52% fewer than scikit-learn and 21% fewer than XGBoost**.
- Implemented a **cost-sensitive threshold optimizer** minimizing missed-fraud cost and false-positive friction cost; the selected operating point covered **99.99% of fraudulent transaction value** at a **0.13% false-positive rate**, reducing **modeled fraud exposure** from **Rs.632 crore to Rs.3.98 lakh** on the held-out test set.
- Engineered a **post-calibration ranking sanity check** comparing calibrated and raw score ordering before deployment; detected an XGBoost failure where early stopping collapsed the model to a single tree and Platt scaling inverted the risk ranking, causing **ROC-AUC to fall from 0.97 to 0.03** and preventing deployment of the faulty model.

Achievements & Certifications

- **Microsoft Innovate Hackathon 2025 — AI Therapist Assistant:** Developed a Generative AI conversational assistant providing 24/7 personalized mental health support, featuring NLP-driven guided journaling and automated emergency routing.
- **NVIDIA:** Getting Started with Accelerated Computing in Modern CUDA C++
- **DeepLearning.AI:** Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization